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ROYAL UNIVERSITY OF PHNOM PENH

Linux System Admin Assignment

Linux Server Admin

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Class: M1

Group: 1

Group Members (Group 1)



Name	Role	Tasks
Sou Chanrojame	Leader	DHCP Server, VPN Server, FTP Server, DNS Server, Failover Cluster, Mail Server, Terminal Server, Fine-Tuning LLM
Orn Pheakdey	Member	File Server, Proxy Server
Long Neron	Member	Database Server, Domain Controller
Then Sivthean	Member	Web Server, Container (Docker)
Le Sreyima	Member	Backup Server, Load Balancing

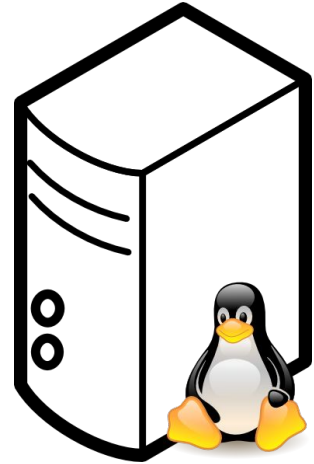
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File Server

Samba is an open-source software suite that enables file and printer sharing between Linux/Unix systems and Windows clients using the SMB/CIFS protocol. It allows a Linux server to function as a file server, providing centralized storage, shared directories, and access control for multiple users across a network. A file server stores and manages files for clients, enabling users to read, write, and collaborate on data securely, while Samba ensures compatibility with Windows environments and seamless integration into mixed-network infrastructures.



Proxy Server

Squid is a caching and forwarding proxy server that handles web requests on behalf of clients, improving performance and controlling access to the internet. By storing frequently accessed content locally, Squid reduces bandwidth usage and speeds up response times for repeated requests. A proxy server acts as an intermediary between client devices and external servers, providing benefits like content filtering, anonymity, security, and traffic management, making it useful for organizations to optimize and regulate internet access.



*Squid
Proxy*

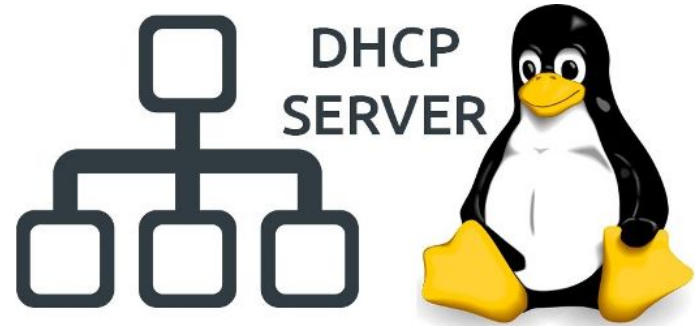
DNS Server

BIND is the most widely used software for implementing a Domain Name System (DNS) server, which translates human-readable domain names into IP addresses that computers use to locate each other on a network. BIND allows administrators to configure authoritative DNS zones, manage domain records, and handle recursive queries, enabling reliable name resolution for both internal networks and the internet. A DNS server, in general, is essential for directing network traffic efficiently, supporting web browsing, email delivery, and other services that rely on domain names.



DHCP Server

A DHCP (Dynamic Host Configuration Protocol) Server is a network service that automatically assigns IP addresses and other critical network configuration parameters (such as subnet masks, default gateways, and DNS server addresses) to devices (clients) when they connect to a network.



VPN Server

A VPN (Virtual Private Network) server acts as a secure, intermediate gateway between a user's device and the internet, establishing an encrypted "tunnel" for all data traffic. It is the cornerstone of privacy-focused browsing and secure remote access.



Terminal Server

TigerVNC is a high-performance, platform-independent implementation of the Virtual Network Computing (VNC) protocol that allows users to remotely access and control graphical desktops on a server. A terminal server provides multiple users with simultaneous access to a centralized server environment, letting them run applications and desktops remotely while all processing happens on the server. Together, TigerVNC enables secure remote graphical sessions, effectively turning a terminal server into an accessible desktop environment for users from anywhere on the network.



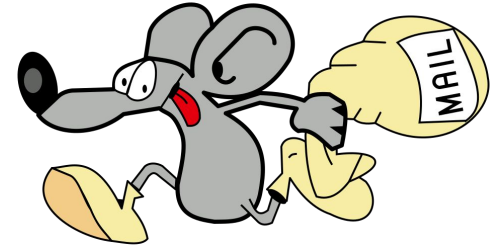
Web Server

Apache HTTP Server, often called Apache httpd, is a widely used open-source web server that delivers web content to clients over the Internet using the HTTP and HTTPS protocols. It handles requests from browsers, serves static and dynamic content, supports modules for features like URL rewriting, authentication, and SSL/TLS encryption, and can host multiple websites on a single server. A web server, in general, is responsible for processing client requests, managing resources, and ensuring reliable delivery of web pages, applications, and services to users.



Mail Server

A mail server is a system that sends, receives, stores, and manages email messages over a network using standard protocols such as SMTP for sending and POP3 or IMAP for retrieving emails. It handles user authentication, message routing, spam filtering, and mailbox storage, enabling reliable communication between individuals and organizations. Mail servers can be configured for internal communication within a company or for public internet email services, and they often include security features like encryption and authentication to protect messages in transit and at rest.

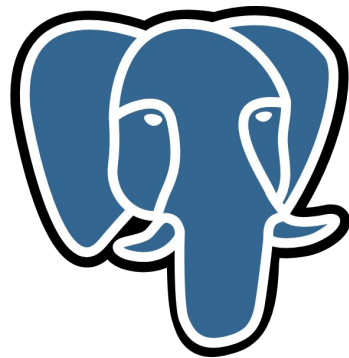


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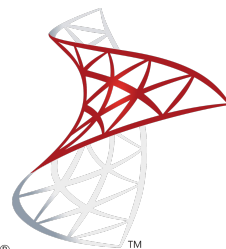


Database Server

PostgreSQL, MongoDB, and Microsoft SQL Server are popular database management systems that store, organize, and manage data for applications. PostgreSQL and SQL Server are relational databases that use structured schemas and SQL queries for reliable transaction processing, while MongoDB is a NoSQL database optimized for flexible, document-oriented storage. A database server provides centralized data management, allowing multiple clients or applications to securely read, write, and manipulate data efficiently, ensuring consistency, availability, and scalability.



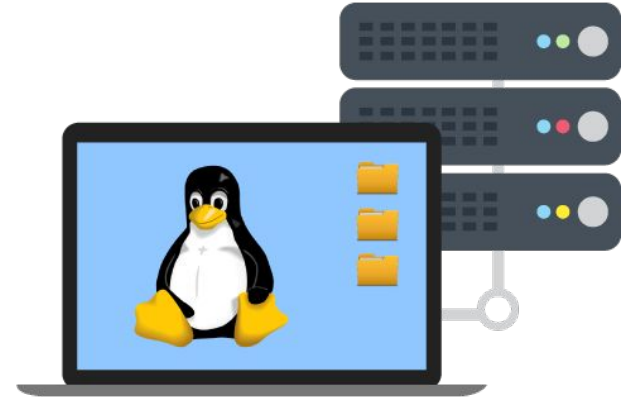
mongoDB®



Microsoft®
SQL Server®

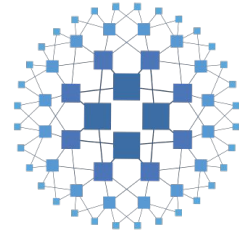
Backup Server

Cronie is a time-based job scheduler for Linux systems that runs background tasks automatically at specified times using crontab configuration files, making it ideal for routine system maintenance like log rotation, updates, and automated backups. A backup server is a dedicated system designed to store copies of data from other machines to protect against data loss caused by hardware failure, human error, cyberattacks, or disasters; it centralizes backup management and enables recovery of files, databases, or entire systems when needed. Together, Cronie can automate scheduled backup tasks, while the backup server securely stores and manages the backup data.



Load Balancing

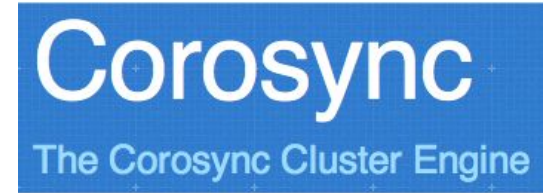
Load balancing is a technique used to distribute incoming network traffic or application requests across multiple servers to ensure no single server becomes overloaded. By spreading workloads evenly, it improves system performance, reliability, and availability, especially in high-traffic environments. Load balancers can operate at different layers of the network (such as transport or application layer) and may use methods like round-robin, least connections, or IP hashing to decide how requests are distributed.



HAPROXY

Failover Cluster

PCS (Pacemaker/Corosync Configuration System) is a command-line tool used to configure and manage high-availability clusters; it provides an easier interface to control cluster components. Pacemaker is the cluster resource manager that decides where and how services (like web servers, databases, or virtual IPs) run, ensuring they fail over automatically if a node goes down. Corosync is the cluster communication engine that keeps nodes connected and synchronized by handling messaging, membership, and quorum. Together, PCS configures the cluster, Corosync manages node communication, and Pacemaker controls resource availability to provide reliable high-availability systems.



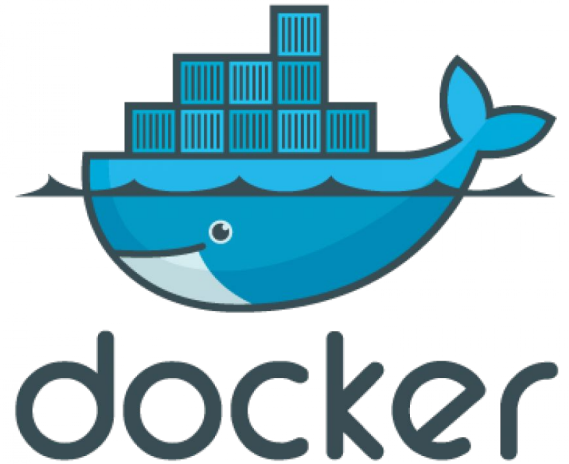
FTP Server

An FTP server is a system that uses the File Transfer Protocol (FTP) to allow users to upload, download, and manage files over a network. It enables file sharing between clients and a central server using authentication with usernames and passwords, and can support both anonymous and secure access configurations. FTP servers are commonly used for website file management, internal file distribution, and data exchange between systems, and can be secured using extensions like FTPS to encrypt data during transmission.



Container

Docker is an open-source containerization platform that allows developers to package applications and their dependencies into lightweight, portable containers that run consistently across different environments. Instead of relying on full virtual machines, Docker uses operating system–level virtualization to isolate applications while sharing the host kernel, making it faster and more resource-efficient. It simplifies development, testing, and deployment workflows by ensuring that applications behave the same way on a developer's laptop, a server, or in the cloud.



Domain Controller

FreeIPA is a free, open-source Identity Management (IdM) solution designed primarily for Linux and Unix-based networked environments. Often described as the Linux equivalent of Microsoft Active Directory, it provides a centralized platform for managing authentication, authorization, and network policies.



